

LODWAR, Kenya

WMO: 636120

Lat: 3.122N

Lon: 35.609E

Elev: 523

StdP: 95.20

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.8	21.8	5.1	5.8	32.9	7.4	6.8	32.3	9.5	28.7	8.5	29.7	1.3	90	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	10.9	37.6	20.8	37.1	20.8	36.5	20.9	24.5	30.6	24.1	30.4	23.7	30.3	5.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.2	19.1	26.3	22.7	18.5	26.3	22.1	17.9	26.4	77.1	30.6	75.4	30.4	74.0	30.0	30.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.3	8.8	7.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			12.3	26.3	1.6	1.6	11.1	27.5	10.2	28.4	9.3	29.3	8.2	30.4	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.8	29.4	30.3	30.7	30.1	30.1	29.7	29.1	29.3	30.2	30.5	29.7	29.0
	DBStd	1.62	1.45	1.42	1.81	2.17	1.71	1.35	1.21	1.20	1.19	1.48	1.71	1.33
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7242	601	569	643	604	622	589	592	600	607	636	589	590
	CDD18.3	4201	342	336	385	354	363	339	334	342	357	377	339	332
	CDH23.3	57153	4714	4933	5691	4901	4981	4465	4167	4344	4882	5275	4467	4333
	CDH26.7	30782	2703	2997	3413	2759	2673	2203	1895	2053	2581	2921	2317	2268
Wind	WSAvg	3.4	2.9	2.8	3.3	3.4	3.5	3.6	3.4	3.5	3.9	4.0	3.4	2.9
Precipitation	PrecAvg	167	9	9	22	51	27	8	15	8	5	10	15	9
	PrecMax	306	68	34	95	189	107	53	132	61	71	89	108	71
	PrecMin	52	0	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	70	16	11	24	46	30	14	29	16	14	21	24	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.2	38.2	38.2	37.8	36.6	35.4	34.7	35.1	36.4	36.7	36.2	36.2
		MCWB	20.0	20.2	21.1	21.9	22.2	21.9	21.4	21.4	21.1	21.3	21.2	20.4
	2%	DB	36.5	37.6	37.6	36.9	35.8	34.8	33.9	34.3	35.6	36.0	35.4	35.5
		MCWB	20.0	20.2	21.1	22.0	22.0	21.7	21.2	21.2	21.0	21.3	21.3	20.6
	5%	DB	36.0	37.0	37.0	36.2	35.2	34.1	33.3	33.8	35.0	35.4	34.9	34.9
		MCWB	20.2	20.2	21.1	22.1	22.0	21.6	21.1	21.1	21.0	21.3	21.3	20.7
	10%	DB	35.3	36.4	36.3	35.3	34.5	33.5	32.8	33.1	34.3	34.8	34.1	34.2
		MCWB	20.1	20.2	21.2	22.2	22.1	21.5	21.1	21.0	21.0	21.3	21.4	20.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.4	24.0	24.2	25.1	25.0	24.2	23.5	23.7	23.7	24.2	24.6	24.3
		MCDB	32.8	32.2	31.0	30.4	30.7	30.4	29.8	29.9	29.6	29.9	30.4	29.7
	2%	WB	23.5	23.0	23.6	24.4	24.2	23.4	22.8	23.0	23.2	23.6	23.8	23.7
		MCDB	30.5	32.1	31.1	30.2	30.5	30.3	29.5	30.3	30.3	29.9	29.6	29.6
	5%	WB	22.7	22.4	23.1	24.0	23.8	23.0	22.4	22.5	22.8	23.1	23.3	23.2
		MCDB	29.4	32.3	31.0	30.2	30.4	30.5	29.5	30.0	30.7	30.0	29.2	29.1
	10%	WB	22.1	21.9	22.7	23.6	23.4	22.6	22.1	22.1	22.3	22.6	22.9	22.7
		MCDB	30.3	32.4	31.1	30.3	30.3	30.4	29.5	29.7	30.8	30.3	29.0	29.1
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.7	10.9	9.4	8.8	8.1	7.8	8.3	8.8	9.0	8.9	10.7
		MCDBR	13.4	13.4	11.8	10.5	9.3	8.5	8.2	8.8	9.3	9.4	9.8	11.8
		MCWBR	3.7	4.0	2.8	2.6	2.3	2.5	2.3	2.5	2.4	2.3	2.3	3.0
	5% WB	MCDBR	10.1	11.7	10.2	9.1	8.7	7.9	7.9	8.4	8.5	9.1	8.6	9.0
		MCWBR	3.2	4.0	2.9	2.9	2.6	2.6	2.4	2.8	2.5	2.5	2.8	2.7
Clear-Sky Solar Irradiance	taub	0.405	0.437	0.449	0.423	0.432	0.500	0.500	0.454	0.426	0.409	0.414	0.410	
	taud	2.326	2.222	2.212	2.336	2.306	2.063	2.058	2.208	2.285	2.358	2.348	2.335	
	Ebn at Noon	918	894	879	885	855	784	789	845	886	908	902	907	
	Edh at Noon	132	149	151	130	130	163	165	146	138	128	128	129	
All-Sky Solar Radiation	RadAvg	6.55	6.77	6.66	6.14	6.28	6.08	6.01	6.40	6.82	6.38	5.95	6.16	
	RadStd	0.32	0.31	0.41	0.41	0.42	0.27	0.30	0.28	0.28	0.27	0.26	0.35	0.37

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page